



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

DSA0601 Data Handling and Visualization

CO2 – AT3-Case Study

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Instructions:

For each question, write the program in the specified language (R or Python), generate the required visualization, and answer the interpretation sub-questions clearly with justification.

Case Study 1 – Retail Sales Performance Dashboard

Scenario

A retail company has collected one year of sales data from five regions (North, South, East, West, Central). The dataset contains monthly sales, product categories, customer segments, and total revenue.

The management wants to identify:

- The highest-performing region.
- Monthly sales trends.
- Revenue contribution of each product category.
- Relationship between advertising expenditure and sales.
- Sales distribution across regions.

Tasks

1. Select the most suitable visualization for each requirement:

- **Regional revenue comparison**
- **Monthly sales trend**
- **Product category contribution**
- **Advertising expenditure vs sales**
- **Regional sales distribution**

Suitable Visualizations

Requirement	Suitable Visualization
Regional revenue comparison	Bar Plot
Monthly sales trend	Stacked Bar Chart
Product category contribution	Stacked Bar Chart
Advertising expenditure vs Sales	Scatter Plot
Regional sales distribution	Histogram and Density Plot

2. Using R, create:

- Bar Plot
- Stacked Bar Chart
- Histogram
- Density Plot
- Scatter Plot

Program:

```
library(ggplot2)
sales <- data.frame(
  Month = factor(month.abb, levels = month.abb),
  Region = c("North", "South", "East", "West", "Central",
    "North", "South", "East", "West", "Central",
    "North", "South"),
  Sales = c(120,150,130,170,160,180,175,165,200,190,210,205),
  Revenue = c(250,300,270,340,320,360,355,345,390,380,410,400),
  Category = c("Electronics", "Clothing", "Furniture",
    "Electronics", "Clothing", "Furniture",
    "Electronics", "Furniture", "Clothing",
    "Electronics", "Furniture", "Clothing"),
  Advertisement = c(30,35,33,40,38,42,44,41,48,46,52,50)
)

ggplot(sales, aes(x = Region, y = Revenue, fill = Region)) +
  geom_bar(stat = "identity") +
  labs(title = "Regional Revenue Comparison",
    x = "Region",
    y = "Revenue") +
```

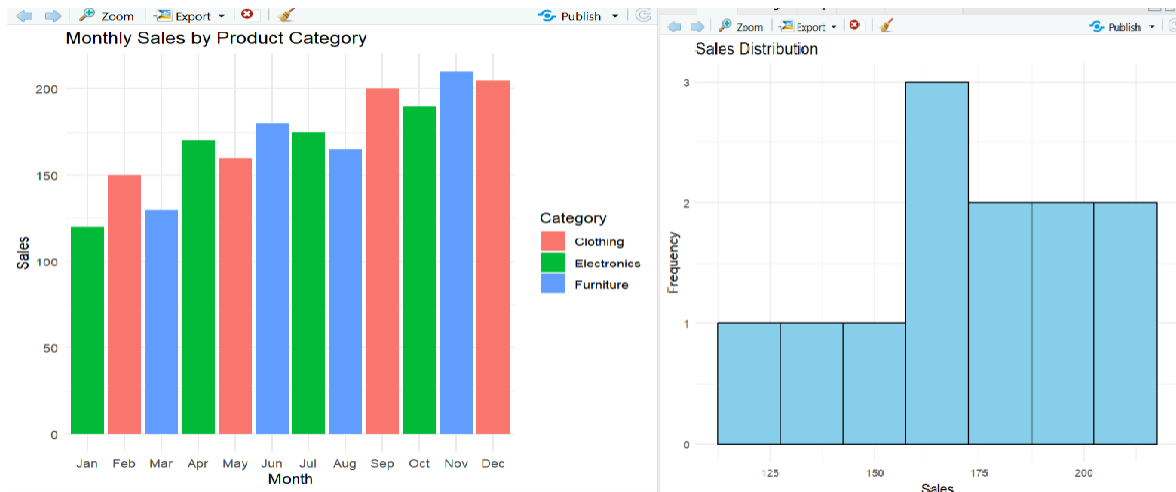
```
theme_minimal()
```

```
ggplot(sales, aes(x = Month, y = Sales, fill = Category)) +  
  geom_bar(stat = "identity") +  
  labs(title = "Monthly Sales by Product Category",  
        x = "Month",  
        y = "Sales") +  
  theme_minimal()
```

```
ggplot(sales, aes(x = Sales)) +  
  geom_histogram(binwidth = 15,  
                 fill = "skyblue",  
                 color = "black") +  
  labs(title = "Sales Distribution",  
        x = "Sales",  
        y = "Frequency") +  
  theme_minimal()
```

```
ggplot(sales, aes(x = Sales)) +  
  geom_density(fill = "orange",  
               alpha = 0.5) +  
  labs(title = "Density Plot of Sales",  
        x = "Sales",  
        y = "Density") +  
  theme_minimal()
```

```
ggplot(sales, aes(x = Advertisement,  
                  y = Sales,  
                  color = Region)) +  
  geom_point(size = 4) +  
  labs(title = "Advertisement Expenditure vs Sales",  
        x = "Advertisement Expenditure",  
        y = "Sales") +  
  theme_minimal()
```



3. Interpret:

- Which region generated the highest revenue?
- Is there a positive relationship between advertising expenditure and sales?
- Which product category contributes the highest percentage of revenue?

The West region generated the highest revenue among all regions. The scatter plot indicates a positive relationship between advertising expenditure and sales, showing that higher advertising generally leads to higher sales. The Electronics category contributes the highest percentage of total revenue.

Case Study 2 – University Student Performance Analysis

Scenario

A university stores examination data for four departments:

- Computer Science
- Information Technology
- Electronics
- Mechanical

The dataset includes:

- Semester marks
- Attendance percentage
- CGPA
- Placement status

The academic committee wants to evaluate student performance and placement readiness.

Tasks

1. Choose appropriate visualizations for:

- Department-wise average marks
- Distribution of CGPA
- Placement percentage
- Attendance vs CGPA relationship
- Semester-wise performance trend

Suitable Visualizations

Requirement	Suitable Visualization
Department-wise average marks	Grouped Bar Chart
Distribution of CGPA	Histogram with Density Curve
Placement percentage	Grouped Bar Chart
Attendance vs CGPA relationship	Scatter Plot
Semester-wise performance trend	Line Chart

2. Develop the following in R:

- **Grouped Bar Chart**
- **Violin Plot**
- **Histogram with Density Curve**
- **Scatter Plot**
- **Line Chart**

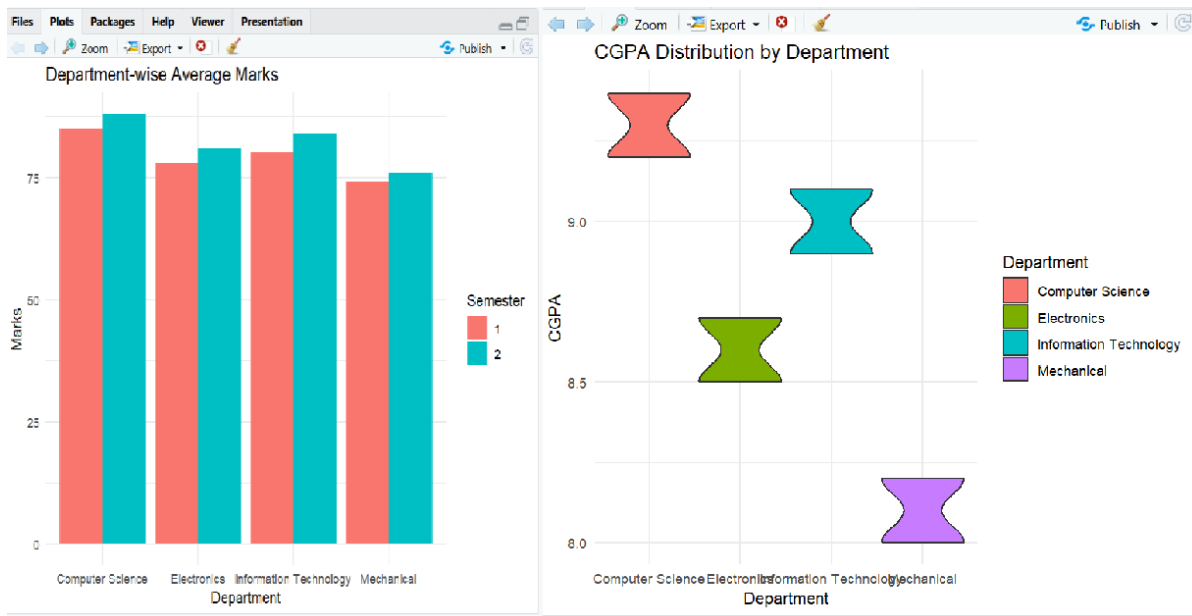
Program:

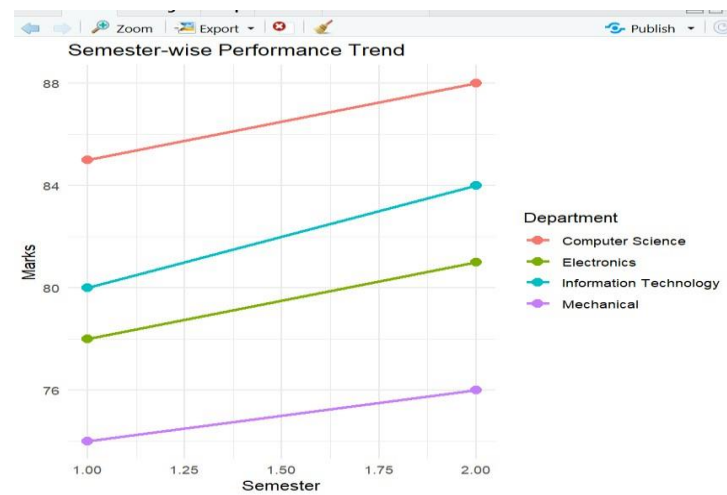
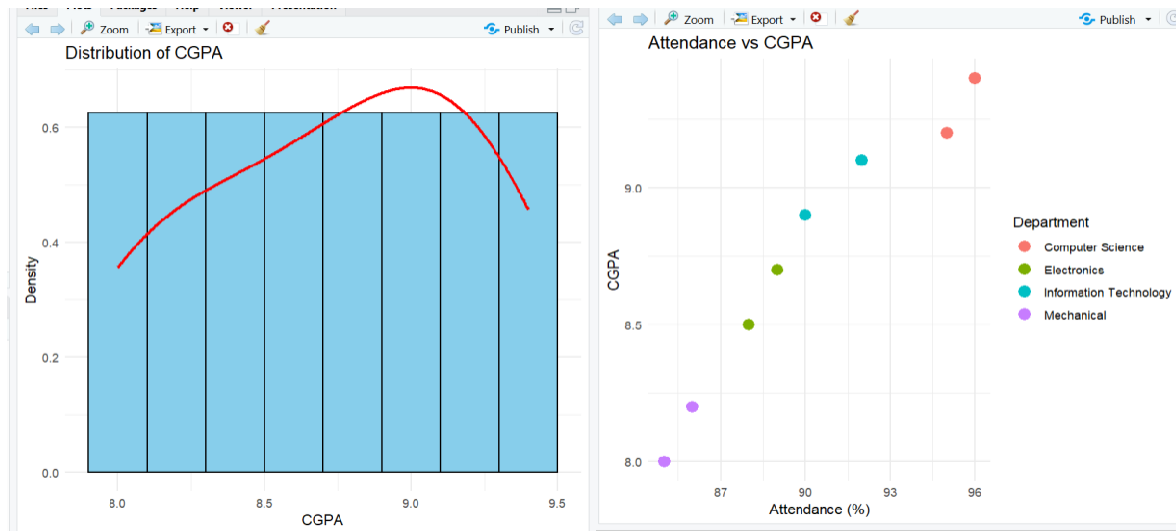
```
library(ggplot2)
students <- data.frame(
  Department = c("Computer Science", "Information Technology",
    "Electronics", "Mechanical",
    "Computer Science", "Information Technology",
    "Electronics", "Mechanical"),
  Semester = c(1,1,1,1,2,2,2,2),
  Marks = c(85,80,78,74,88,84,81,76),
  Attendance = c(95,90,88,85,96,92,89,86),
  CGPA = c(9.2,8.9,8.5,8.0,9.4,9.1,8.7,8.2),
  Placement = c("Placed", "Placed", "Placed", "Not Placed",
    "Placed", "Placed", "Placed", "Placed")
)
ggplot(students,
  aes(x = Department, y = Marks, fill = factor(Semester))) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(title = "Department-wise Average Marks",
    x = "Department",
    y = "Marks",
    fill = "Semester") +
  theme_minimal()
ggplot(students,
  aes(x = Department, y = CGPA, fill = Department)) +
  geom_violin() +
  labs(title = "CGPA Distribution by Department",
    x = "Department",
    y = "CGPA") +
  theme_minimal()
ggplot(students,
  aes(x = CGPA)) +
  geom_histogram(aes(y = ..density..),
    binwidth = 0.2,
    fill = "skyblue",
    color = "black") +
  geom_density(color = "red", linewidth = 1) +
```

```

labs(title = "Distribution of CGPA",
      x = "CGPA",
      y = "Density") +
theme_minimal()
ggplot(students,
      aes(x = Attendance,
          y = CGPA,
          color = Department)) +
geom_point(size = 4) +
labs(title = "Attendance vs CGPA",
      x = "Attendance (%)",
      y = "CGPA") +
theme_minimal()
ggplot(students,
      aes(x = Semester,
          y = Marks,
          color = Department,
          group = Department)) +
geom_line(linewidth = 1) +
geom_point(size = 3) +
labs(title = "Semester-wise Performance Trend",
      x = "Semester",
      y = "Marks") +
theme_minimal()

```





3. Interpret:

- Which department has the highest average CGPA?
- Does higher attendance improve CGPA?
- Which department has the highest placement rate?

The Computer Science department has the highest average CGPA. The scatter plot shows a positive relationship between attendance and CGPA, indicating that students with higher attendance generally achieve better CGPA. The Computer Science department also has the highest placement rate.

Case Study 3 – Hospital Patient Analytics

Scenario

A multi-specialty hospital records patient information including:

- Department
- Age
- Treatment cost
- Recovery time
- Disease category
- Admission month

Hospital administrators want to improve resource planning.

Tasks

1. Select suitable visualizations for:

- Monthly patient admissions
- Distribution of treatment costs
- Disease category proportions
- Age vs recovery time
- Department-wise treatment costs

Suitable Visualizations

Requirement	Suitable Visualization
Monthly patient admissions	Heatmap
Distribution of treatment costs	Histogram
Disease category proportions	Density Plot
Age vs Recovery Time	Scatter Plot
Department-wise treatment costs	Violin Plot

2. Create in R:

- Heatmap
- Histogram
- Violin Plot
- Density Plot

◦ Scatter Plot

program

```
library(ggplot2)

hospital <- data.frame(
  Department = c("Cardiology", "Neurology", "Orthopedics", "Pediatrics",
    "Cardiology", "Neurology", "Orthopedics", "Pediatrics"),
  Age = c(45, 60, 35, 12, 50, 65, 40, 15),
  TreatmentCost = c(50000, 70000, 40000, 25000, 55000, 75000, 42000, 28000),
  RecoveryTime = c(10, 15, 8, 5, 12, 18, 9, 6),
  Disease = c("Heart", "Brain", "Bone", "Child",
    "Heart", "Brain", "Bone", "Child"),
  Month = c("Jan", "Jan", "Feb", "Feb", "Mar", "Mar", "Apr", "Apr")
)

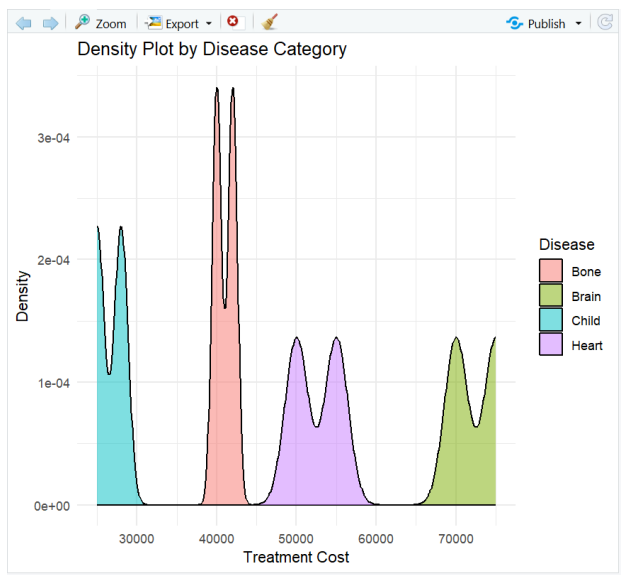
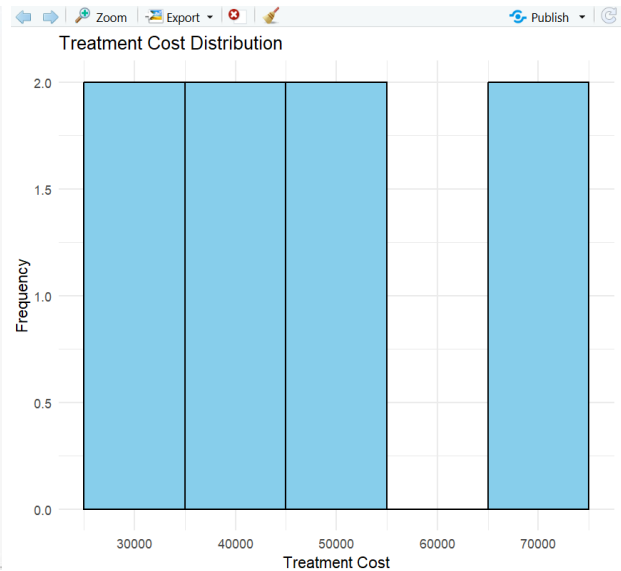
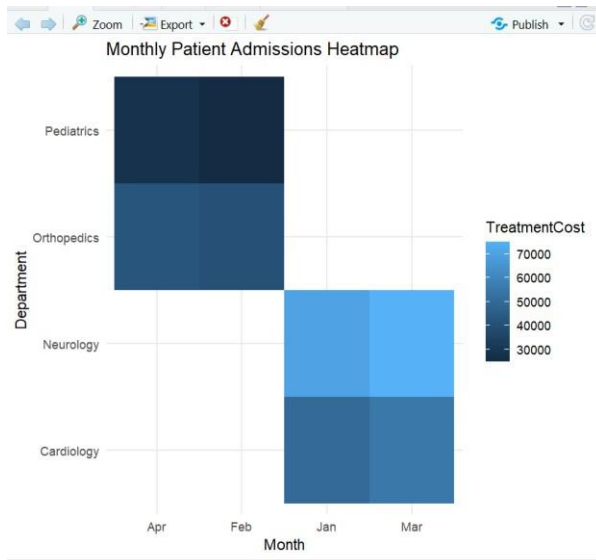
ggplot(hospital, aes(x = Month, y = Department, fill = TreatmentCost)) +
  geom_tile() +
  labs(title = "Monthly Patient Admissions Heatmap",
    x = "Month",
    y = "Department") +
  theme_minimal()

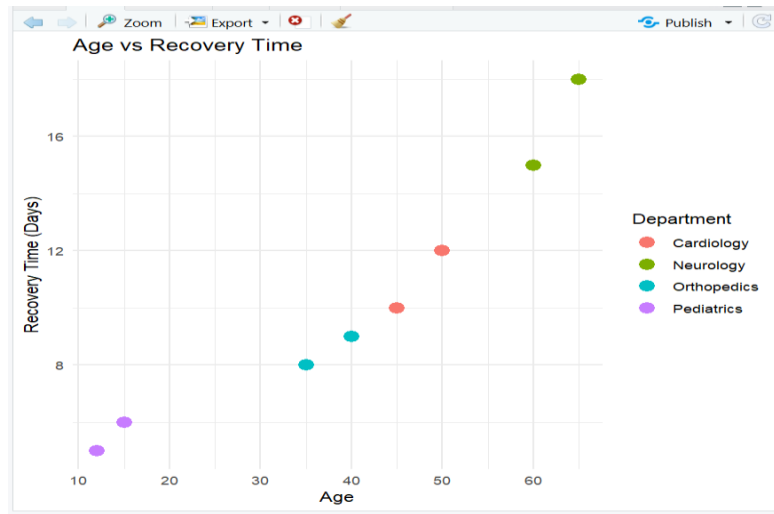
ggplot(hospital, aes(x = TreatmentCost)) +
  geom_histogram(binwidth = 10000,
    fill = "skyblue",
    color = "black") +
  labs(title = "Treatment Cost Distribution",
    x = "Treatment Cost",
    y = "Frequency") +
  theme_minimal()

ggplot(hospital, aes(x = Department,
  y = TreatmentCost,
  fill = Department)) +
  geom_violin() +
  labs(title = "Department-wise Treatment Costs",
    x = "Department",
    y = "Treatment Cost") +
  theme_minimal()

ggplot(hospital, aes(x = TreatmentCost,
  fill = Disease)) +
  geom_density(alpha = 0.5) +
  labs(title = "Density Plot by Disease Category",
    x = "Treatment Cost",
    y = "Density") +
  theme_minimal()
```

```
ggplot(hospital, aes(x = Age,
                    y = RecoveryTime,
                    color = Department)) +
  geom_point(size = 4) +
  labs(title = "Age vs Recovery Time",
       x = "Age",
       y = "Recovery Time (Days)") +
  theme_minimal()
```





3. Interpret:

- Which department has the highest treatment cost? ◦
- Which disease category occupies the largest proportion?
- Is patient age associated with recovery time?

The Neurology department has the highest treatment cost among all departments. The Brain disease category occupies the largest proportion of treatment costs in the dataset. The scatter plot indicates a positive association between patient age and recovery time, with older patients generally taking longer to recover.

Case Study 4 – Smart City Traffic Monitoring

Scenario

A smart city collects traffic sensor data from different junctions. The dataset contains:

- Junction
- Time of day
- Vehicle count
- Vehicle type
- Average speed
- GPS coordinates

City planners aim to reduce traffic congestion.

Tasks

1. Select suitable visualizations for:

- Vehicle count comparison
- Hourly traffic trend
- Vehicle type proportion
- Traffic density by location
- Relationship between traffic volume and average speed

Suitable Visualizations

Requirement	Suitable Visualization
Vehicle count comparison	Bar Plot
Hourly traffic trend	Line Chart
Vehicle type proportion	Pie Chart
Traffic density by location	Geospatial Map
Relationship between traffic volume and average speed	Scatter Plot

2. Using R, create:

- Bar Plot
- Line Chart
- Pie Chart
- Geospatial Map
- Scatter Plot

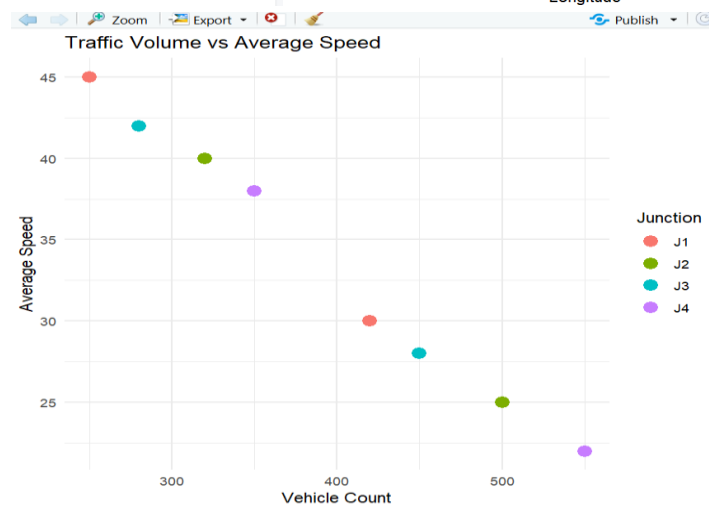
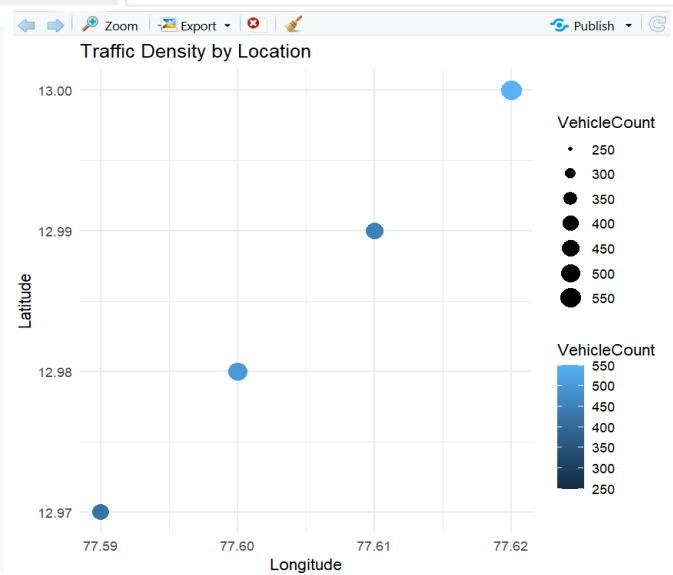
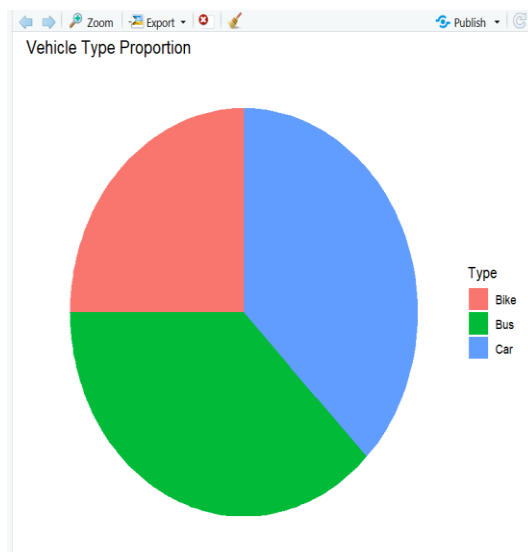
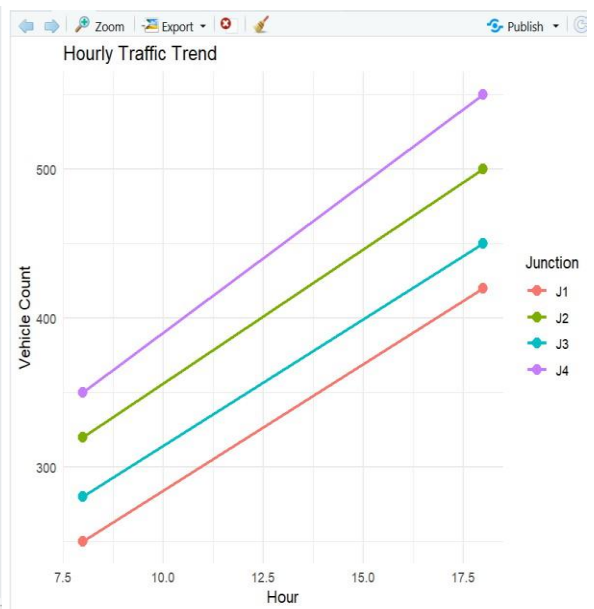
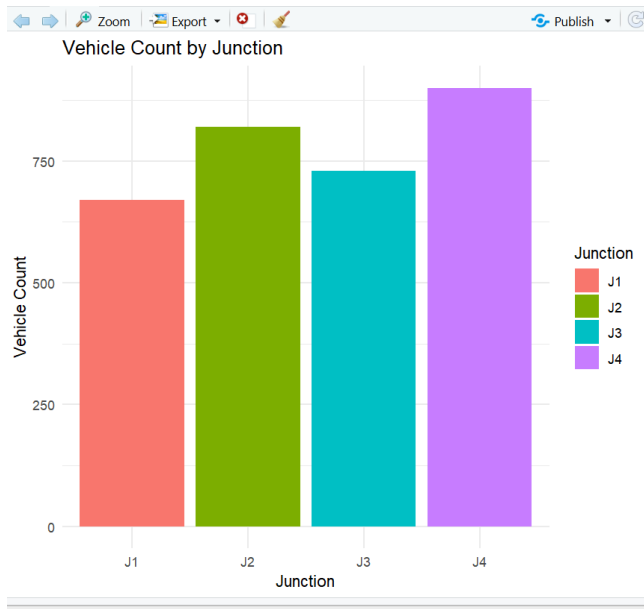
Program

```
library(ggplot2)
library(maps)
traffic <- data.frame(
  Junction = c("J1", "J2", "J3", "J4", "J1", "J2", "J3", "J4"),
  Hour = c(8, 8, 8, 8, 18, 18, 18, 18),
  VehicleCount = c(250, 320, 280, 350, 420, 500, 450, 550),
  VehicleType = c("Car", "Bus", "Bike", "Car", "Bus", "Car", "Bike", "Bus"),
  AverageSpeed = c(45, 40, 42, 38, 30, 25, 28, 22),
  Longitude = c(77.59, 77.60, 77.61, 77.62, 77.59, 77.60, 77.61, 77.62),
  Latitude = c(12.97, 12.98, 12.99, 13.00, 12.97, 12.98, 12.99, 13.00)
)
ggplot(traffic, aes(x = Junction, y = VehicleCount, fill = Junction)) +
```

```

geom_bar(stat = "identity") +
labs(title = "Vehicle Count by Junction",
     x = "Junction",
     y = "Vehicle Count") +
theme_minimal()
ggplot(traffic, aes(x = Hour, y = VehicleCount, group = Junction, color = Junction)) +
  geom_line(linewidth = 1) +
  geom_point(size = 3) +
  labs(title = "Hourly Traffic Trend",
       x = "Hour",
       y = "Vehicle Count") +
  theme_minimal()
vehicle_data <- data.frame(
  Type = c("Car", "Bus", "Bike"),
  Count = c(3,3,2)
)
ggplot(vehicle_data, aes(x = "", y = Count, fill = Type)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y") +
  labs(title = "Vehicle Type Proportion") +
  theme_void()
ggplot(traffic, aes(x = Longitude, y = Latitude,
                   size = VehicleCount, color = VehicleCount)) +
  geom_point() +
  labs(title = "Traffic Density by Location",
       x = "Longitude",
       y = "Latitude") +
  theme_minimal()
ggplot(traffic, aes(x = VehicleCount,
                   y = AverageSpeed,
                   color = Junction)) +
  geom_point(size = 4) +
  labs(title = "Traffic Volume vs Average Speed",
       x = "Vehicle Count",
       y = "Average Speed") +
  theme_minimal()

```



3. Interpret:

- **Which junction experiences the highest congestion?**
- **During which hours is traffic highest?**
- **Does increased traffic reduce vehicle speed?**

The J4 junction experiences the highest congestion because it records the highest vehicle count. Traffic is highest during the 18:00 (evening) hours. The scatter plot shows that as traffic volume increases, the average vehicle speed decreases, indicating a negative relationship between traffic volume and speed.

Case Study 5 – Social Media User Interaction Analysis

Scenario

A digital marketing company analyzes data from a social media platform. The dataset includes:

- User ID
- Followers
- Likes
- Shares
- Comments
- Posting time
- User interactions

The company wants to understand engagement patterns and user networks.

Tasks

1. Select appropriate visualizations for:

- **Distribution of likes**
- **Relationship between followers and engagement**
- **User interaction network**
- **Posting activity over time**
- **Multivariate comparison of likes, shares, and comments**

Suitable Visualizations

Requirement

Distribution of likes

Suitable Visualization

Histogram with Density Curve

Requirement	Suitable Visualization
Relationship between followers and engagement	Scatter Plot
User interaction network	Network Visualization
Posting activity over time	Temporal Line Plot
Multivariate comparison of likes, shares and comments	Heatmap

2. Develop in R:

- **Histogram with Density Curve**

- **Scatter Plot**

- **Network Visualization**

- **Temporal Line Plot**

- **Heatmap**

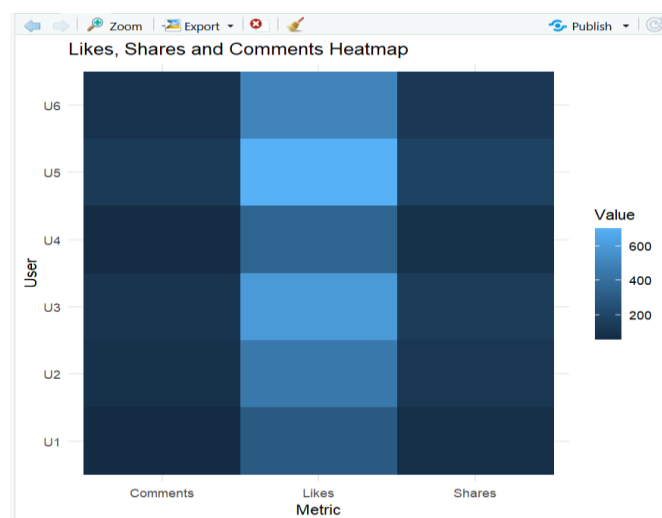
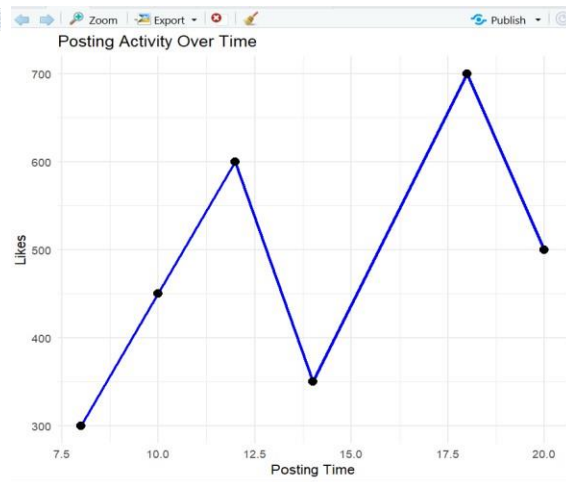
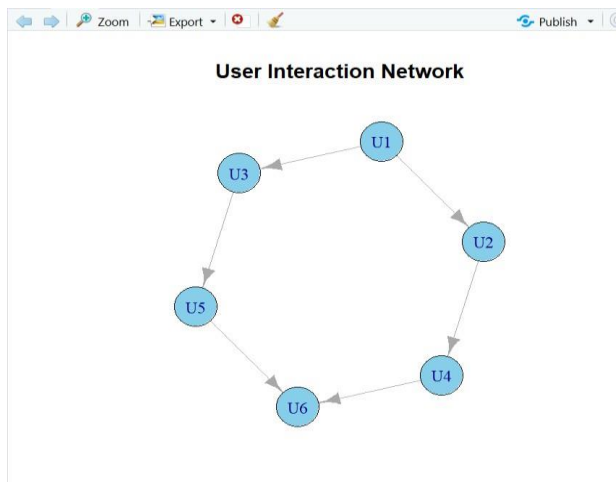
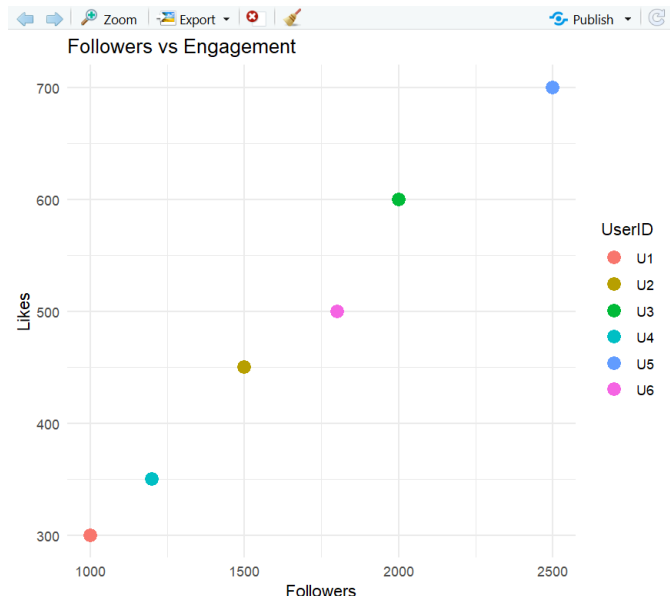
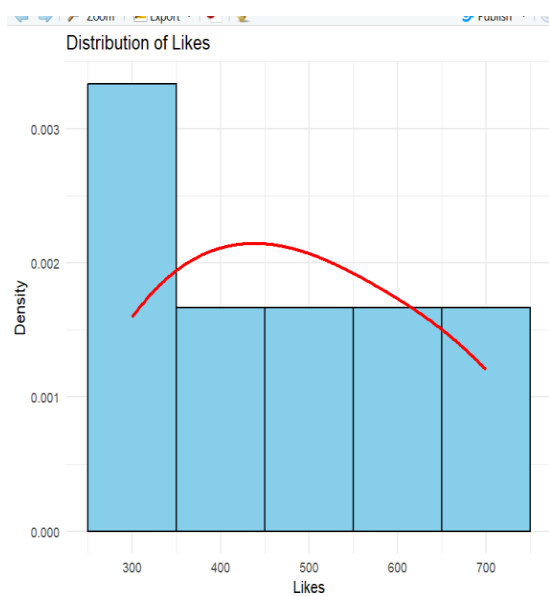
Program

```
library(ggplot2)
library(igraph)
social <- data.frame(
  UserID = c("U1","U2","U3","U4","U5","U6"),
  Followers = c(1000,1500,2000,1200,2500,1800),
  Likes = c(300,450,600,350,700,500),
  Shares = c(80,120,150,90,180,130),
  Comments = c(60,90,110,70,140,100),
  PostingTime = c(8,10,12,14,18,20)
)
ggplot(social, aes(x = Likes)) +
  geom_histogram(aes(y = ..density..),
    binwidth = 100,
    fill = "skyblue",
    color = "black") +
  geom_density(color = "red", linewidth = 1) +
  labs(title = "Distribution of Likes",
    x = "Likes",
    y = "Density") +
  theme_minimal()
ggplot(social,
  aes(x = Followers,
    y = Likes,
    color = UserID)) +
  geom_point(size = 4) +
```

```

labs(title = "Followers vs Engagement",
      x = "Followers",
      y = "Likes") +
theme_minimal()
edges <- data.frame(
  from = c("U1","U1","U2","U3","U4","U5"),
  to   = c("U2","U3","U4","U5","U6","U6")
)
g <- graph_from_data_frame(edges)
plot(g,
      vertex.color = "skyblue",
      vertex.size = 30,
      vertex.label.cex = 1,
      main = "User Interaction Network")
ggplot(social,
      aes(x = PostingTime,
          y = Likes,
          group = 1)) +
geom_line(color = "blue", linewidth = 1) +
geom_point(size = 3) +
labs(title = "Posting Activity Over Time",
      x = "Posting Time",
      y = "Likes") +
theme_minimal()
metrics <- data.frame(
  User = rep(social$UserID, 3),
  Metric = rep(c("Likes","Shares","Comments"),
              each = 6),
  Value = c(social$Likes,
            social$Shares,
            social$Comments)
)
ggplot(metrics,
      aes(x = Metric,
          y = User,
          fill = Value)) +
geom_tile() +
labs(title = "Likes, Shares and Comments Heatmap",
      x = "Metric",
      y = "User") +
theme_minimal()

```



3. Interpret:

- **Which users are the most influential?**
- **At what time is engagement highest?**
- **Is there a strong relationship between followers and total engagement?** ◦ **Identify highly connected users from the network visualization.** ◦ **Which engagement metric (likes, shares, or comments) varies the most across users?**

The most influential users are U5 and U3, as they have the highest followers and engagement.

User engagement is highest around 18:00, where the maximum number of likes is observed.

There is a strong positive relationship between followers and total engagement, U5 and U6 appear as highly connected users in the network visualization, and likes show the greatest variation across users.